

Credit scoring: understanding the trade-offs

Dataset pre-validation proposal

For group discussion and instructor review by 24 September, 09:40

The decision we want to study

A hypothetical lender needs a score that distinguishes good from bad credit outcomes. We propose comparing logistic regression, Random Forest and a tabular foundation model to determine what is gained or lost in predictive accuracy, explainability, stability and fairness. The final recommendation will identify the most defensible model and the conditions required before any real deployment.

Proposed data

South German Credit, corrected UCI version. The public dataset contains 1,000 historical credit records, 20 predictors and a binary outcome. The source file and its coding table are included in the accompanying folder. The original target is `kredit=1` for good credit and 0 for bad credit; modelling will use `y_bad=1-kredit`.

Check completed	Result
Size and outcome	1,000 records; 700 good and 300 bad outcomes
Integrity	No incomplete records, duplicate rows or invalid category codes
Age coverage	19 to 75 years; 150 records below 25 and 850 aged 25 or above
Availability	Public data, CC BY 4.0; source attribution included

Why it fits the assignment

The binary target supports all three model families. Numeric and categorical predictors allow global and individual explanations. Repeated training runs can expose unstable scores or explanations. Age provides a clearly observed attribute for comparing decisions and error rates across groups. A simulated error-cost framework supplies the economic comparison.

Limits to state upfront

The records date from 1973-1975 and bad outcomes were oversampled. Sample probabilities cannot be treated as current population default rates. The amount variable was transformed, so economic results will use illustrative cost units. Sex cannot reliably be recovered from the combined personal-status field; we will not construct male/female groups from it. Valid source categories such as unknown/no savings do not guarantee complete information.

Source: South German Credit [Dataset] (2020), UCI Machine Learning Repository. doi.org/10.24432/C5QG88. Original file SHA-256 and complete coding table are included in the reproducibility package.

Analysis plan and readiness

Proposed protocol. No model results are claimed at this stage.

Dimension	Planned comparison
Performance	ROC-AUC, average precision, log-loss, Brier score and calibration. Choose a decision threshold on validation data using explicit error-cost ratios; report costs per 100 cases.
Interpretability	Logistic coefficients and odds ratios; permutation importance for the three models; local explanations on the same credit profiles, with their assumptions stated.
Stability	Repeat fits with different training samples and seeds. Compare probability variation, decision flips and importance rankings on fixed evaluation cases.
Fairness	By age group: acceptance rate, refusal rate among good outcomes and acceptance rate among bad outcomes. Report counts, rate differences and uncertainty. Compare models with and without age.

Evaluation rules

Use one reproducible 60/20/20 train/validation/test split, stratified jointly by outcome and the primary age group. Fit encoders and scaling on training data only. Treat coded categories as categories. Tune models and decision thresholds on validation data; reserve the test set for the final frozen comparison. Never include kredit, a row identifier or the audit group as an input. All models must use the same test cases.

For the economic analysis, accept a credit if p_{bad} is below the selected threshold. Compare illustrative bad-acceptance to good-refusal cost ratios of 1, 3 and 5. Costs and calibration describe this sample, not a real bank portfolio. Fairness intervals will be conditional on the fitted models; training variation is assessed separately.

Age groups: feasibility checked

Primary group	Records	Bad outcomes	Observed bad rate
Age below 25	150	62	41.3%
Age 25 or above	850	238	28.0%

The 25-year cut-off is an analytical choice, not a legal threshold. Repeat the analysis at 30 (369 / 631 records). A cut-off at 20 is unusable for inference: only two records fall below it. These descriptive outcome differences alone do not establish discrimination. Test-set subgroup estimates will remain imprecise.

What needs agreement before Thursday

Confirm the six-person team and its dataset choice; request instructor pre-validation; verify access, licence and compute requirements for a specific TabPFN version before committing to that implementation. Record the package and checkpoint versions. The public benchmark may overlap prior model evaluations, so any generalisation claim needs caution.

Already completed: source integrity, coding checks, target validation, age-group counts and sensitivity audit. **Next:** implement and compare the three models, build the interactive application and prepare the 15-minute presentation. Final slides, complete code and application are due by email to perignon@hec.fr on 28 September at 09:40.

Method reference: github.com/PriorLabs/TabPFN. Assignment source: ISAF Project Description, HEC Paris, 2026-2027. Instructor approval and group adoption are pending.